

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) Total Marks: 100

ANNEXURE A

Assignment 1

Code of Conduct:

I N . H e m a n t h R e d d y (192472215) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Questions

Annexure A

Assignment 1

Question 34 – Campus Bus Tracking

Problem Statement:

A university tracks buses and routes.

Questions

- a) Develop a Java program using **HashMap**.
- b) Search buses by route number.
- c) Compare Map implementations.
- d) Explain smart transportation.

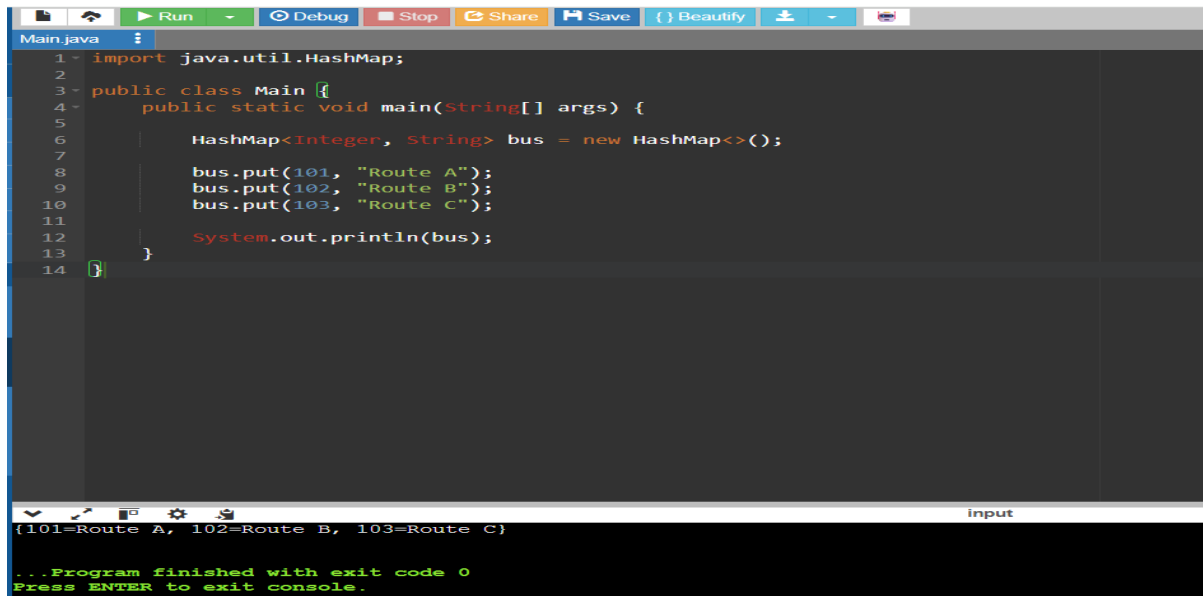
(a) Develop a Java program using HashMap

```
import java.util.HashMap;
```

```
public class Main {  
    public static void main(String[] args) {  
  
        HashMap<Integer, String> bus = new HashMap<>();  
  
        bus.put(101, "Route A");  
        bus.put(102, "Route B");  
        bus.put(103, "Route C");  
  
        System.out.println(bus);  
    }  
}
```

Output:

Questions



The screenshot shows an IDE window titled 'Main.java' with the following Java code:

```
1 import java.util.HashMap;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         HashMap<Integer, String> bus = new HashMap<>();
7
8         bus.put(101, "Route A");
9         bus.put(102, "Route B");
10        bus.put(103, "Route C");
11
12        System.out.println(bus);
13    }
14 }
```

The console output at the bottom shows:

```
{101=Route A, 102=Route B, 103=Route C}
...Program finished with exit code 0
Press ENTER to exit console.
```

(b) Search Bus by Route Number

```
import java.util.HashMap;
```

```
public class Main {
    public static void main(String[] args) {
```

```
        HashMap<Integer, String> bus = new HashMap<>();
```

```
        bus.put(101, "Route A");
```

```
        bus.put(102, "Route B");
```

```
        bus.put(103, "Route C");
```

```
        int route = 102;
```

```
        if (bus.containsKey(route))
```

```
            System.out.println(bus.get(route));
```

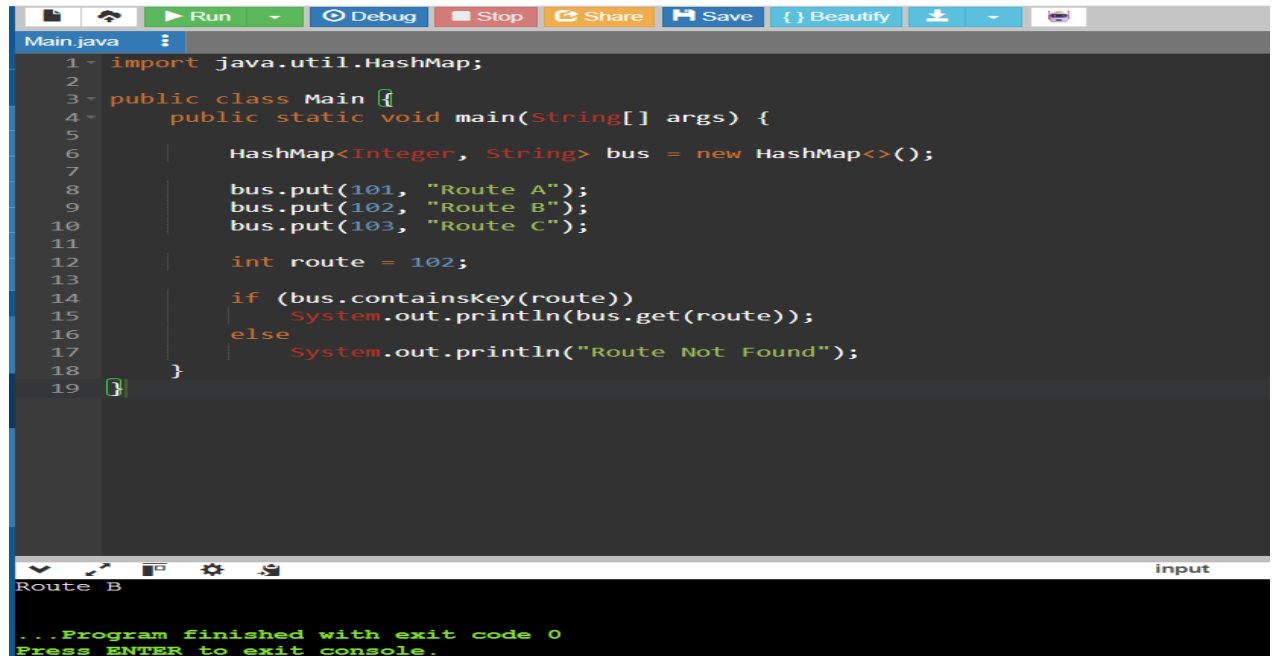
```
        else
```

```
            System.out.println("Route Not Found");
```

```
    }
```

Questions

}



```
1 import java.util.HashMap;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         HashMap<Integer, String> bus = new HashMap<>();
7
8         bus.put(101, "Route A");
9         bus.put(102, "Route B");
10        bus.put(103, "Route C");
11
12        int route = 102;
13
14        if (bus.containsKey(route))
15            System.out.println(bus.get(route));
16        else
17            System.out.println("Route Not Found");
18    }
19 }
```

Route B

...Program finished with exit code 0
Press ENTER to exit console.

((c) Compare Map Implementations

Map	Features
HashMap	Fast, does not maintain order
LinkedHashMap	Maintains insertion order
TreeMap	Stores keys in sorted order

(c) Explain Smart Transportation

Smart transportation uses technologies such as GPS, sensors, mobile applications, and the Internet to improve public transport. It helps in real-time bus tracking, reduces waiting time, improves passenger safety, and provides efficient route management.

Advantages:

- Real-time bus tracking
- Reduced travel time
- Improved safety
- Better traffic management
- Easy route monitoring